

BOLSHENARYMSKOE, KazakhstanWMO: **364280**Lat: **49.200N**Lon: **84.517E**Elev: **1316**StdP: **14.01**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.3	-28.1	-38.0	0.7	-33.2	-33.0	0.9	-27.9	18.4	20.7	14.1	18.2	2.3	30	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	90.1	66.2	87.0	65.4	83.9	64.4	69.3	83.6	67.7	81.7	66.2	80.0	6.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.5	95.5	75.0	62.8	89.8	73.8	61.2	84.9	72.4	34.3	83.7	32.9	81.6	31.7	80.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	14.3	11.4	DB	-36.0	96.0	6.4	3.3	-40.6	98.4	-44.3	100.3	-47.9	102.1	-52.5	104.5	(4)
(5)			WB	-36.5	73.4	6.4	4.1	-41.1	76.3	-44.9	78.7	-48.5	81.0	-53.2	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	-5.0	0.2	18.2	42.8	56.1	66.7	70.0	66.7	55.3	41.9	23.9	5.7		
	DBStd	28.03	13.58	12.26	13.55	9.79	7.59	6.39	5.26	6.10	7.82	8.30	12.15	12.73		
	HDD50	6823	1706	1394	987	248	30	0	0	0	34	268	784	1372		
	HDD65	10604	2171	1814	1452	665	288	57	16	52	303	715	1234	1837		
	CDD50	2102	0	0	0	32	220	501	619	518	193	19	0	0		
	CDD65	407	0	0	0	0	12	108	170	105	12	0	0	0		
	CDH74	4964	0	0	0	8	232	1204	1804	1424	287	5	0	0		
	CDH80	1665	0	0	0	0	48	380	651	515	70	1	0	0		
	WSAvg	4.2	2.6	2.6	3.3	5.0	5.9	5.1	4.8	5.0	4.8	4.4	3.8	3.0		
	PrecAvg	16.1	0.8	0.6	0.8	1.3	1.9	1.9	2.1	1.7	1.5	1.5	1.3	1.0		
	PrecMax	23.8	1.8	1.2	1.7	3.4	5.6	4.2	7.2	4.9	4.7	3.7	2.8	2.2		
(15) Precipitation	PrecMin	9.9	0.2	0.1	0.3	0.2	0.2	0.4	0.7	0.5	0.1	0.4	0.2	0.4		
	PrecStd	3.1	0.4	0.3	0.4	0.7	1.2	0.9	1.3	1.0	1.1	0.7	0.7	0.4		
	DB	29.3	35.3	53.2	74.6	85.6	91.5	94.4	94.9	87.1	70.8	51.7	38.2			
	MCWB	27.7	32.3	42.6	56.4	63.3	66.1	68.9	66.3	60.8	53.0	42.5	33.6			
	2% DB	24.5	27.3	44.7	69.2	79.9	87.7	90.0	89.8	81.1	65.6	47.0	31.9			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	23.2	25.2	38.5	52.7	59.6	65.6	67.3	65.6	59.3	50.8	40.7	29.9			
	5% DB	20.6	23.0	39.5	64.7	75.8	84.4	86.7	85.8	76.9	60.8	42.8	27.2			
	MCWB	19.4	21.3	35.8	50.4	57.7	65.0	66.7	64.1	57.8	48.8	37.8	25.5			
	10% DB	16.4	19.2	36.1	59.9	71.8	81.1	83.6	82.0	72.4	56.3	38.8	23.9			
	MCWB	15.2	17.7	33.3	48.1	56.1	63.8	65.5	63.1	56.1	46.7	35.2	22.4			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	27.9	32.2	43.7	57.8	64.8	70.7	72.8	69.6	63.8	54.6	44.8	34.1			
	MCDB	29.0	35.1	51.1	71.8	81.8	84.7	86.9	86.8	82.9	66.9	48.5	36.5			
	2% WB	23.2	25.4	39.2	53.9	61.9	68.3	70.2	67.4	61.0	51.8	41.3	29.7			
	MCDB	24.3	27.3	44.0	66.2	75.9	82.9	84.2	83.9	77.6	63.8	46.6	31.9			
	5% WB	19.5	21.4	35.9	51.3	59.6	66.6	68.4	65.8	58.9	49.6	38.1	25.6			
(33) Mean Daily Temperature Range	MCDB	20.6	22.8	39.1	62.5	72.5	80.4	81.6	82.0	74.6	59.1	42.4	27.1			
	10% WB	15.2	17.7	33.6	48.8	57.6	65.0	67.0	64.3	56.7	47.3	35.4	22.4			
	MCDB	16.4	19.3	35.9	58.7	69.2	77.8	79.7	79.2	70.4	55.4	38.6	23.8			
	5% DB	18.1	21.3	20.8	22.4	23.9	23.1	22.9	25.9	26.2	20.7	16.4	16.5			
	MCDBR	16.5	20.9	20.2	31.8	30.2	28.3	28.3	32.3	33.1	29.3	20.2	16.7			
(40) Clear-Sky Solar Irradiance	MCWBR	15.5	18.9	15.1	17.4	14.5	11.7	11.1	13.1	16.2	16.8	14.1	14.6			
	5% WB	16.6	20.9	19.1	28.1	25.9	24.6	24.5	28.2	30.7	26.0	18.8	16.4			
	MCDBR	15.6	19.1	14.4	16.0	13.4	11.1	10.7	12.2	15.8	15.4	13.5	14.5			
	taub	0.282	0.314	0.363	0.411	0.398	0.400	0.413	0.384	0.346	0.342	0.313	0.285			
	taud	2.116	2.063	2.038	2.107	2.225	2.299	2.279	2.332	2.394	2.296	2.165	2.082			
(42) All-Sky Solar Radiation	Ebn at Noon	237	257	263	264	273	273	267	270	270	247	221	214			
	Edh at Noon	26	36	44	46	43	40	40	36	31	28	25	24			
	RadAvg	522	882	1327	1608	1901	2066	1928	1780	1333	764	454	388			
(45)	RadStd	34	31	106	140	148	161	141	108	128	80	44	35			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page